CS & IT ENGINEERING

DIGITAL LOGIC



Logic Gate

Lecture No. 5



By - CHANDAN SIR

Recap of Previous Lecture



Topic

NOT, AND, OR, NAND, NOR

Topics to be Covered

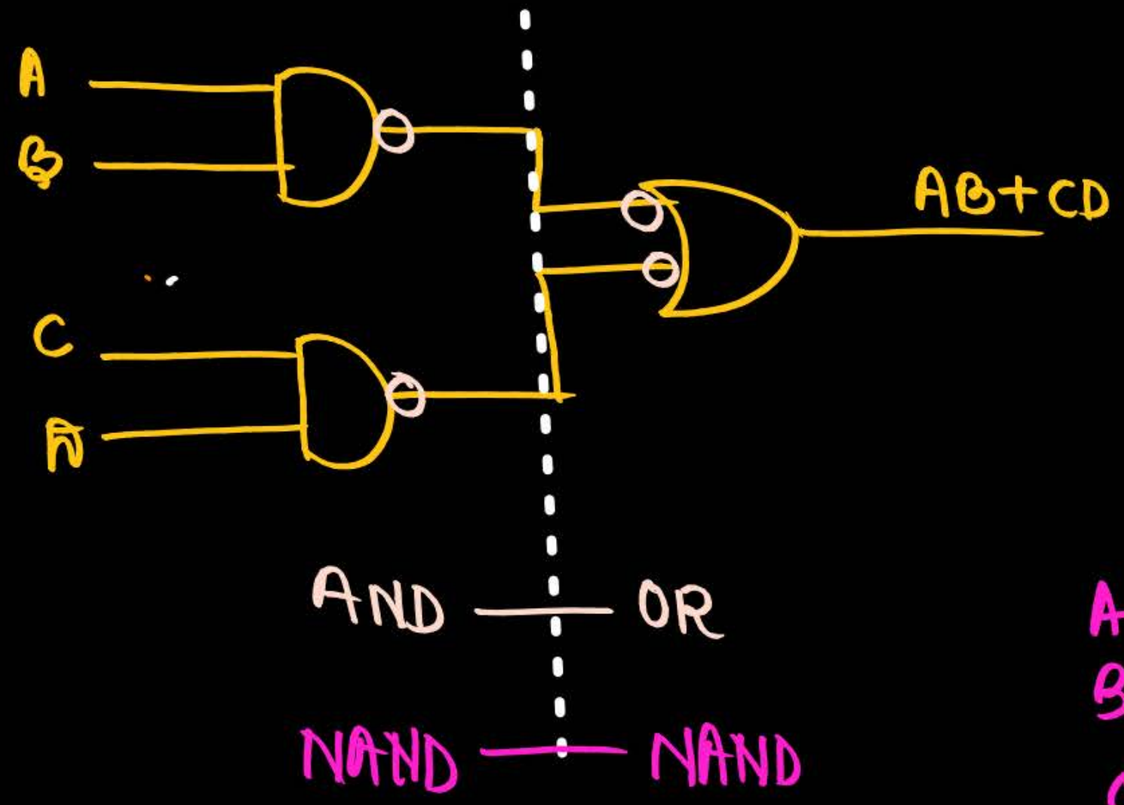


- XOR GATE
- Questions
- Discussion

$A \cdot B \cdot \bar{C} \cdot D \cdot E \cdot$	$A + \bar{B} + C + \bar{D} + E + \ldots$
$NAND = (2n-2) + k$	$NAND = (3n-3) - k$
$NOR = (3n-3) - k$	$NOR = (2n-2) + k$

case(3) $f = AB + CD$ ← SOP

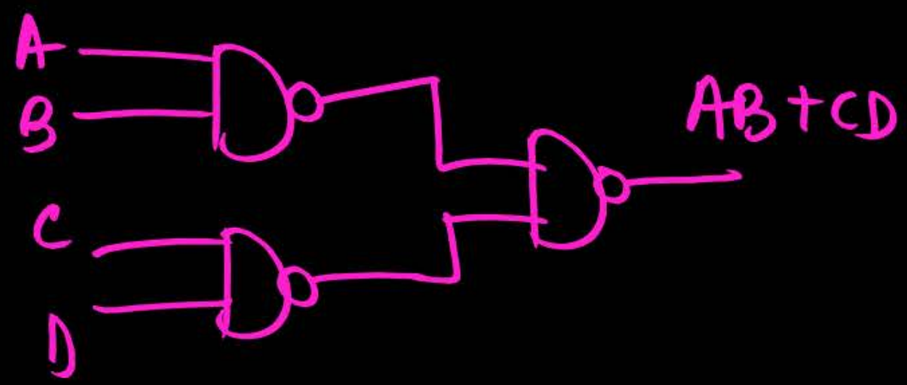
NAND = ?



SOP → Sum of Product

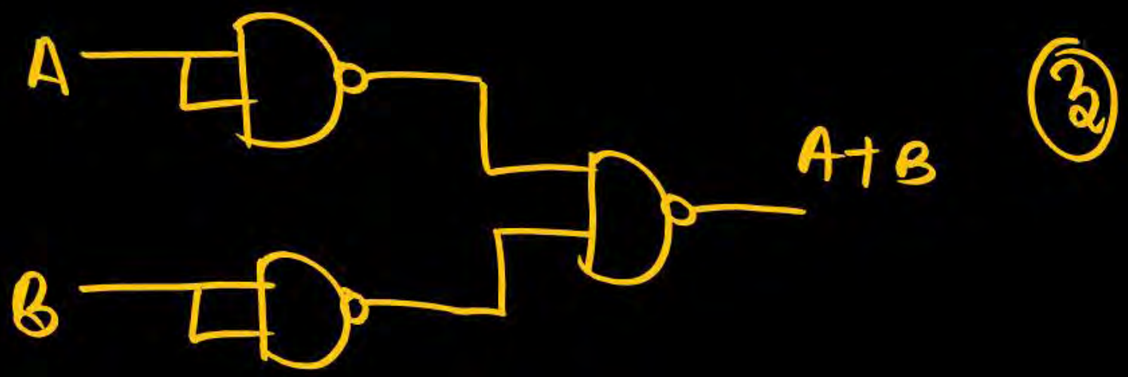
POS → Product of sum

AB+CD



A+B ⟶ OR GATE

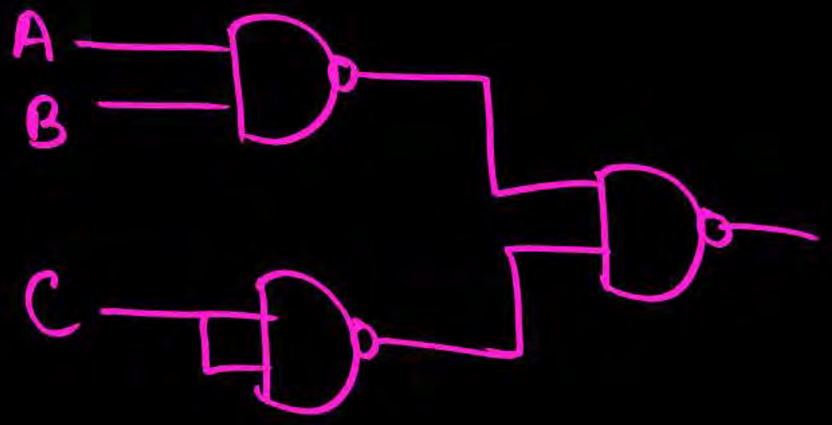
A·A+B·B



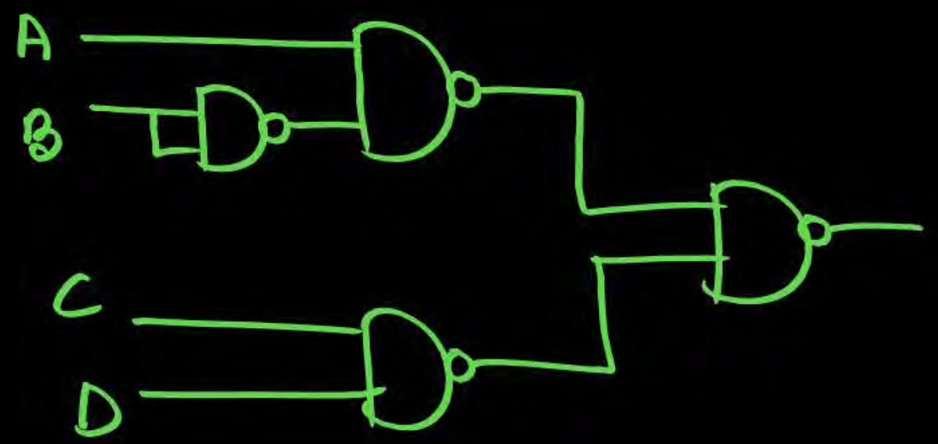
(2)

Q $f = AB + C$ NAND = ?

$AB + C \cdot C$



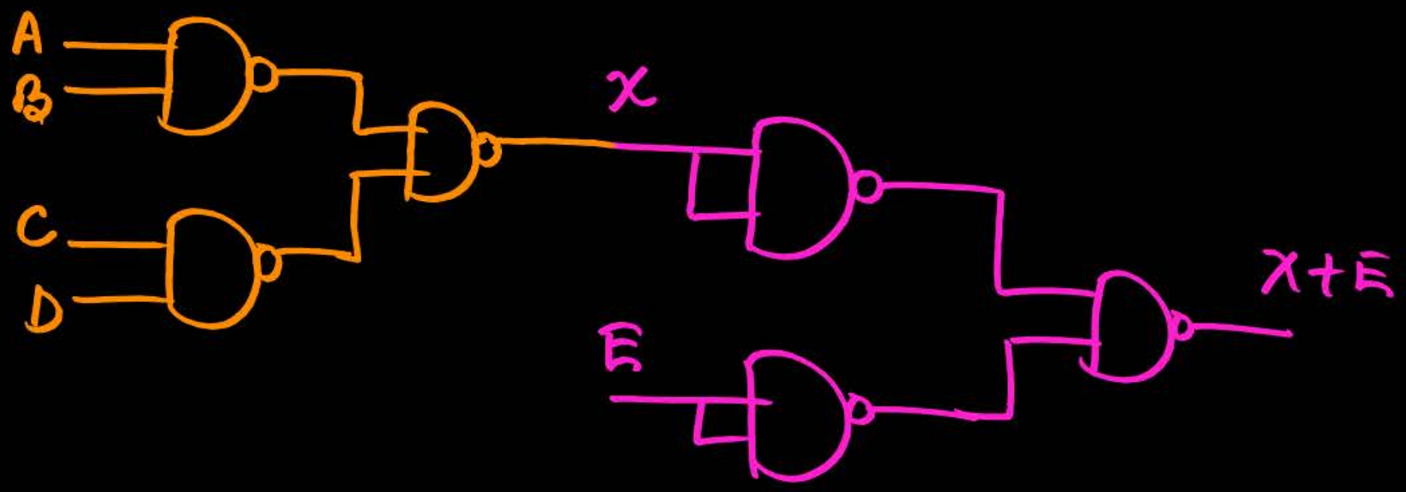
Q $A\bar{B}+CD$ NAND = ?



Q $\underbrace{AB+CD}_{X}+E$ NAND = ?

$X+E$

$X \cdot X + E \cdot E$

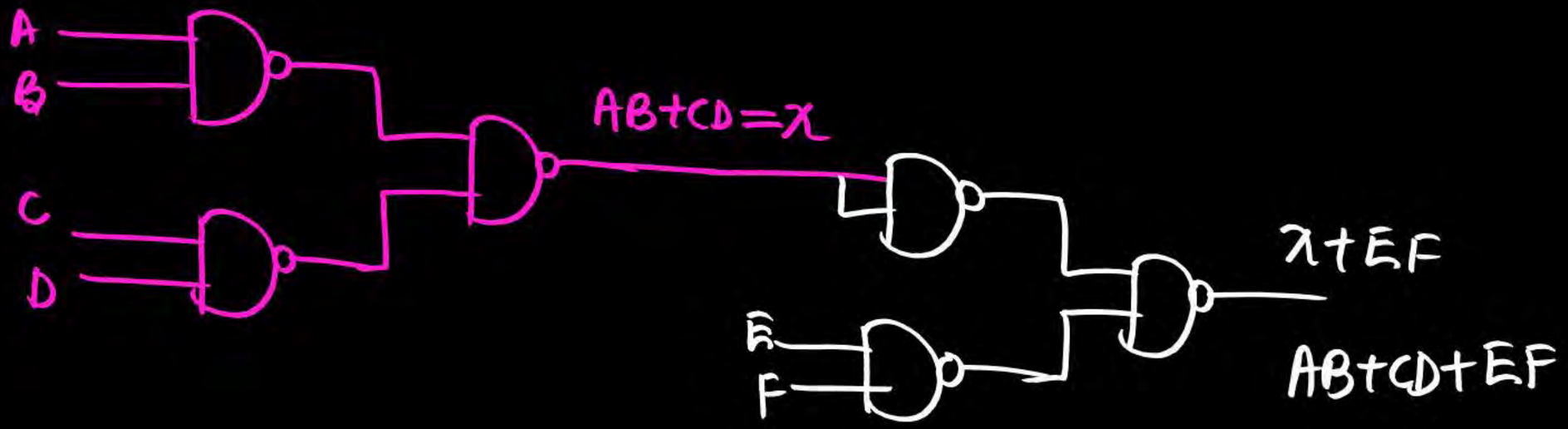


6

Q = AB + CD + EF

$x + EF$

$x \cdot x + EF$



6 =

Q $AB + BC + CD + DA$

NAND = ?

(A) 6

(B) 7

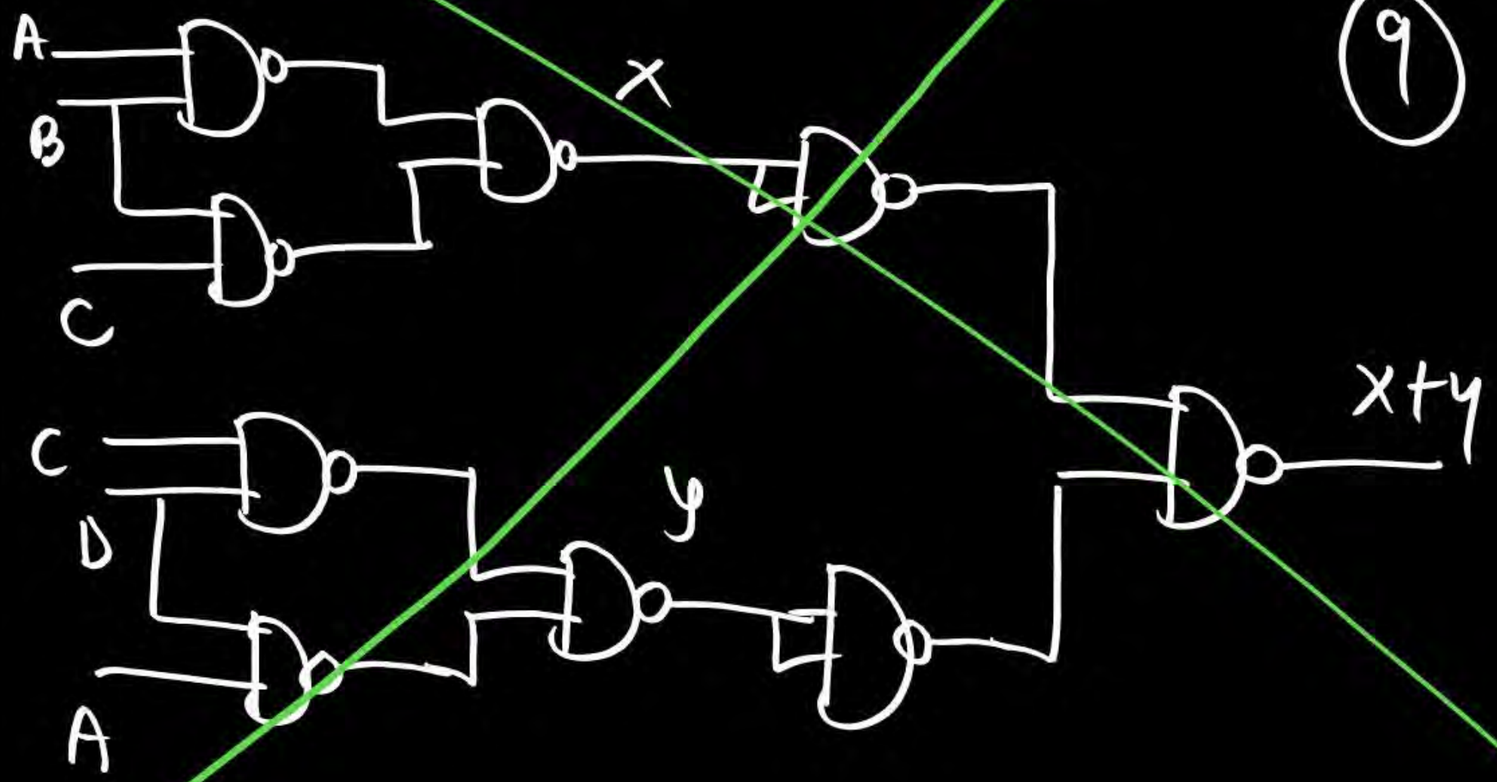
(C) 8

(D) 9

$AB+BC+CD+DA$

x y

$x+y = x \cdot x + y \cdot y$



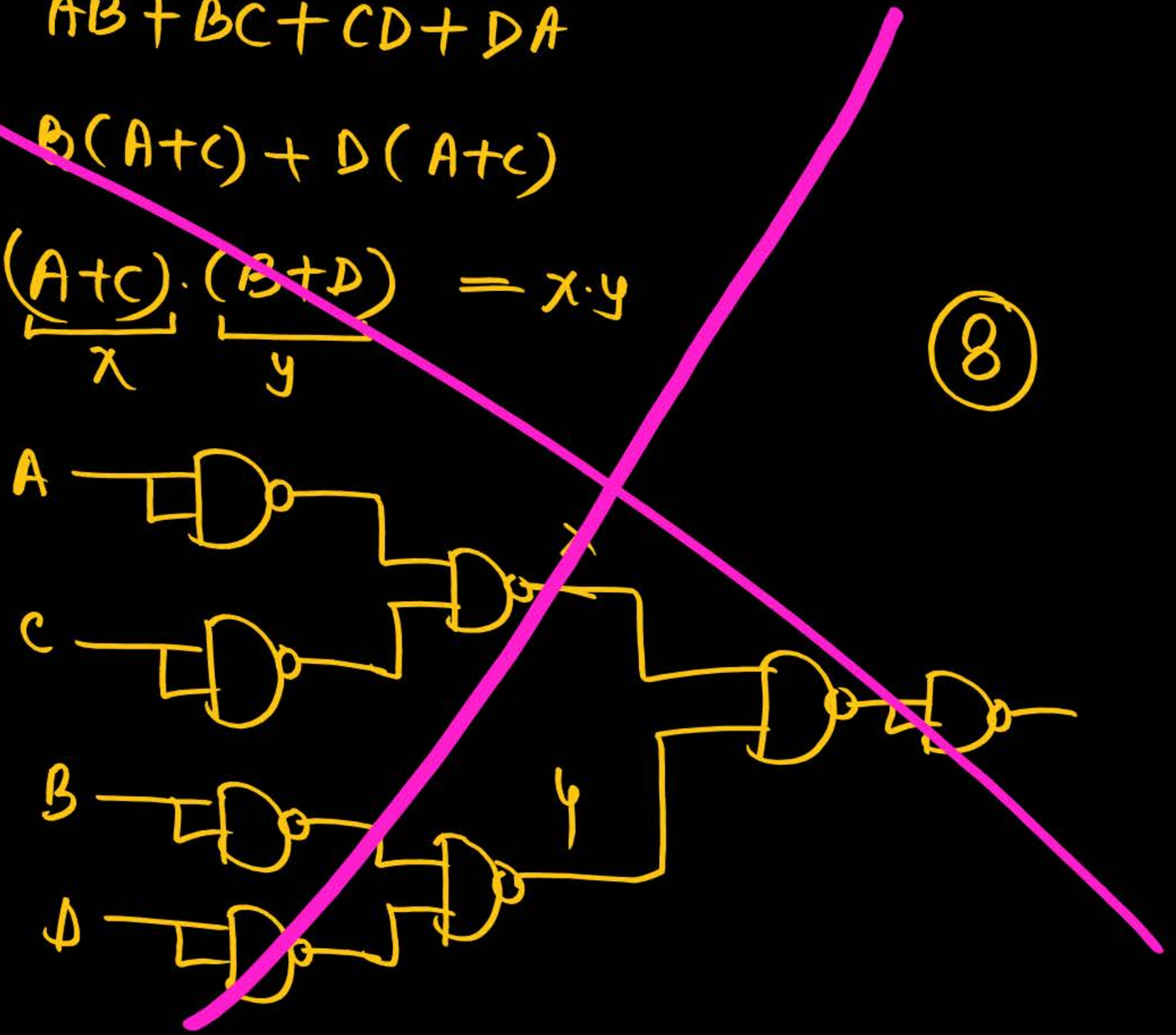
(9)

$AB + BC + CD + DA$

$B(A+C) + D(A+C)$

$\underbrace{(A+C)}_{x} \cdot \underbrace{(B+D)}_{y} = x \cdot y$

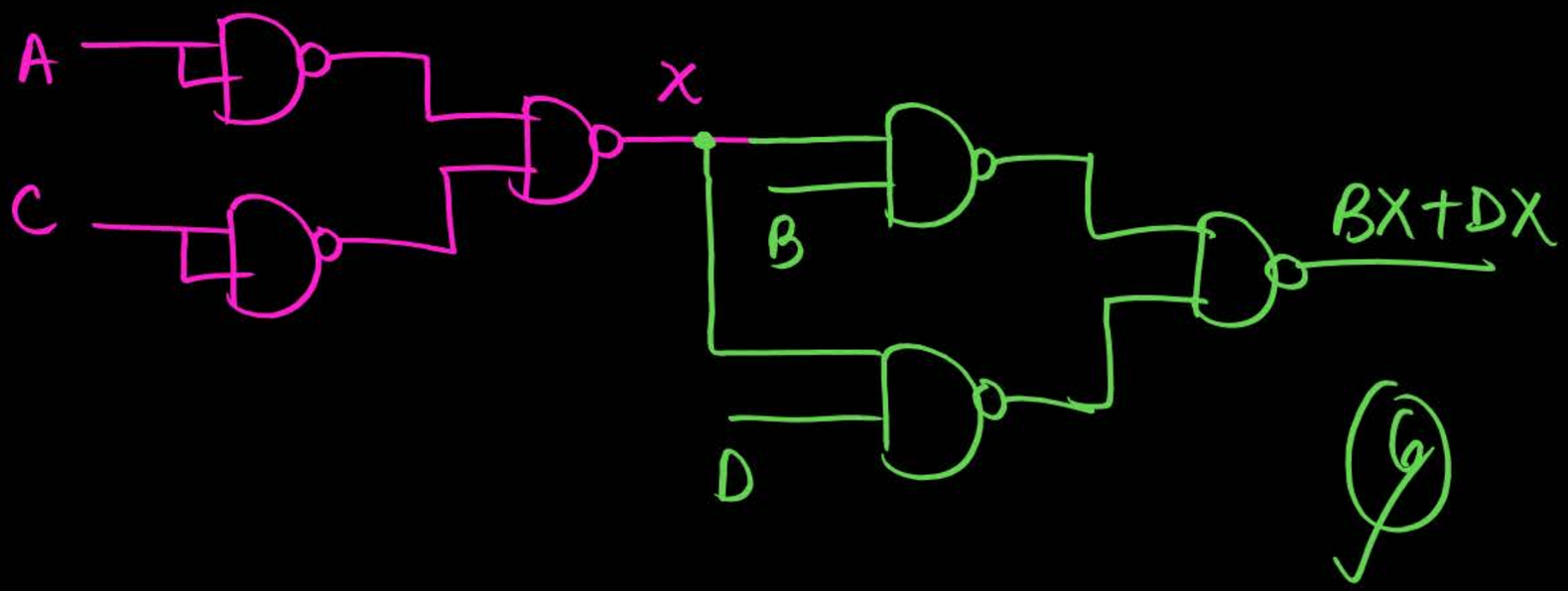
(8)



$AB+BC+CD+DA$

$B(A+C)+D(A+C)$ $\qquad A+C=X$

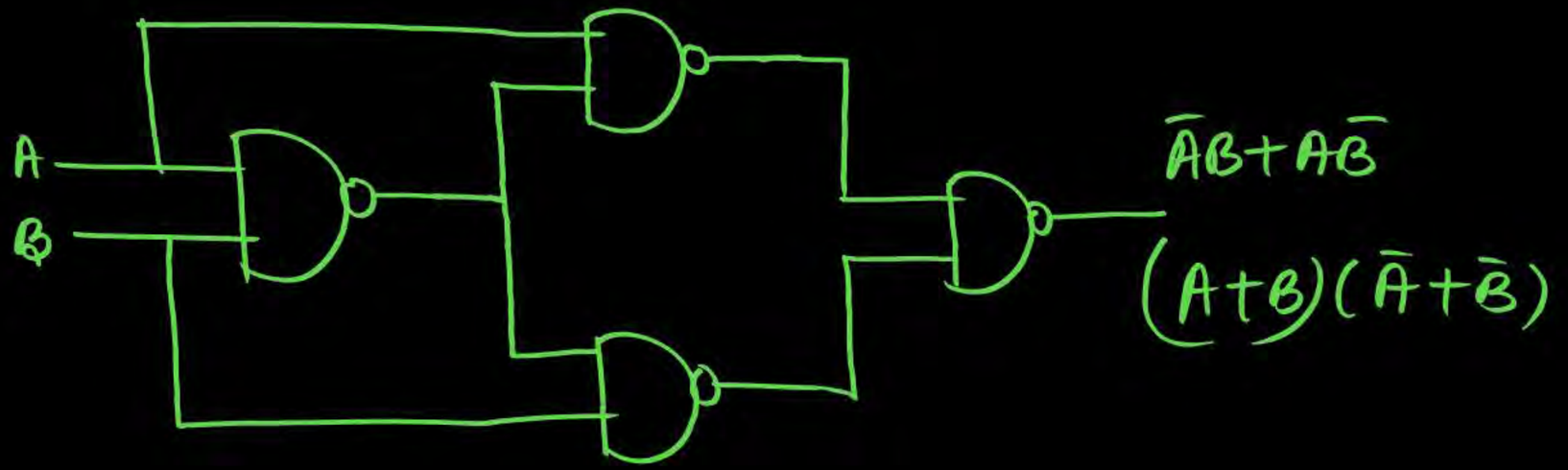
$BX+DX$



Case (4) $\bar{A}B + A\bar{B}$

$(A+B)(\bar{A}+\bar{B})$

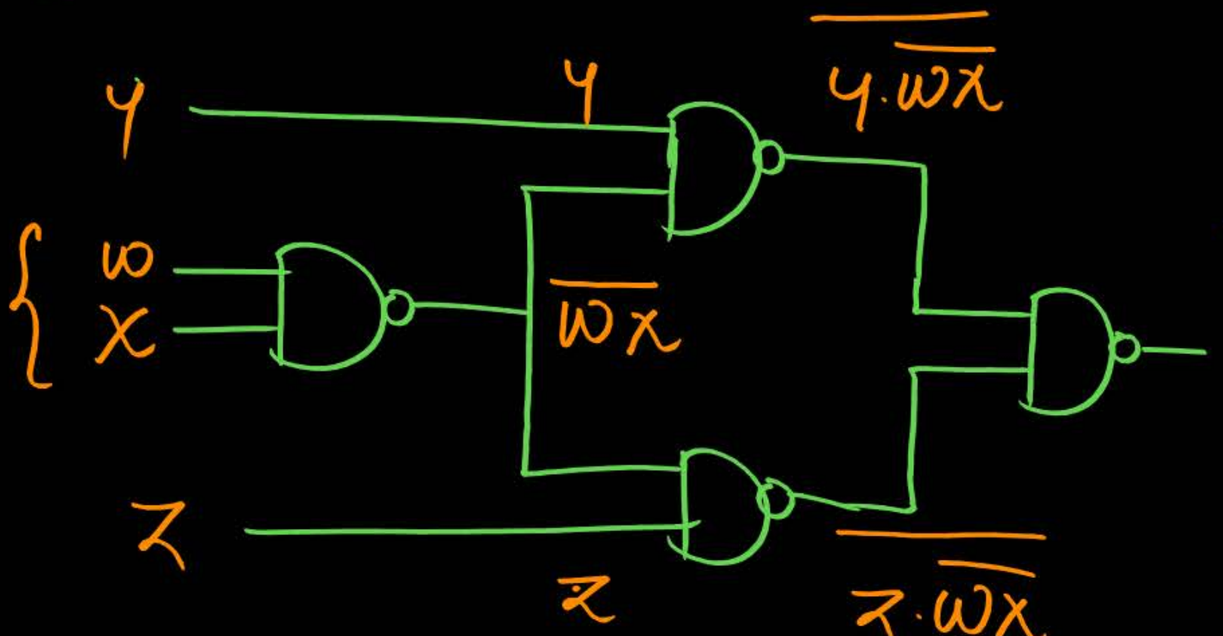
④ NAND



④

(4)

$(\bar{w}+\bar{x})(y+z)$



$\overline{\overline{y\cdot\overline{wx}}\cdot\overline{z\cdot\overline{wx}}}$

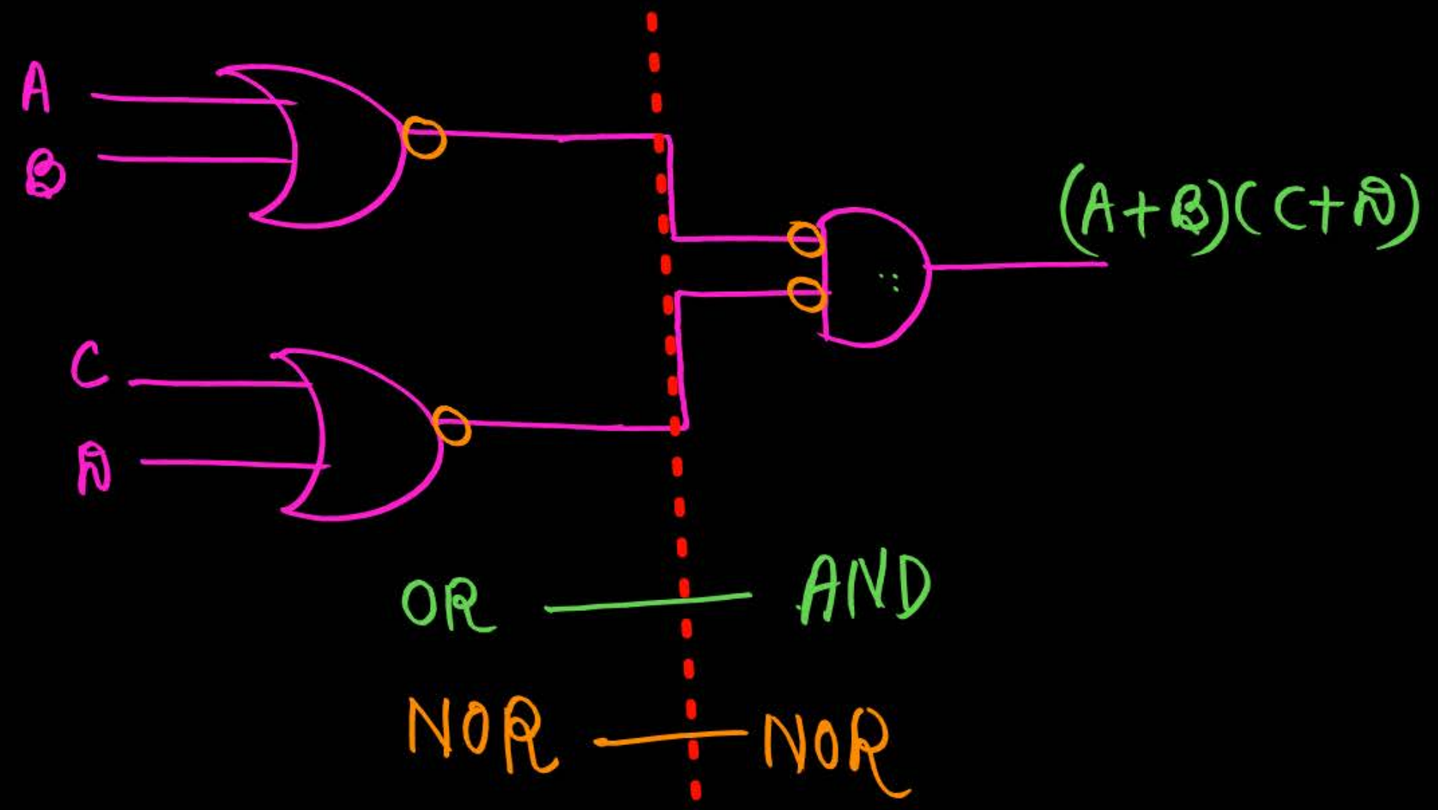
$y\cdot\overline{wx}+z\cdot\overline{wx}$

$\bar{w}x(y+z)$

$(\bar{w}+\bar{x})(y+z)$

$(A+B)\cdot(C+D)$ → POS

NOR



(3)

SOP ⟶ AND – OR

NAND – NAND

POS ⟶ OR – AND

NOR – NOR



Topic : XOR GATE

- **XOR GATE**

1. Symbol



2. Truth Table

A	B	Y = A ⊕ B
0	0	0
0	1	1
1	0	1
1	1	0

③ $A = B \quad y = 0$

$A = \overline{B} \quad y = 1$

④ $A \oplus A = 0$

$A \oplus 0 = A$

$A \oplus \overline{A} = 1$

$A \oplus 1 = \overline{A}$

Ex If $A \oplus B = C$

$B \oplus C = ?$

$B \oplus C = A$

$A \oplus C = B$

$A \oplus B \oplus C =$



⑤ $A \oplus A = 0$

$A \oplus A \oplus A = A$

$A \oplus A \oplus A \oplus A = 0$

$$A \oplus A \oplus A \cdots\cdots \oplus A = \begin{cases} 0 & \text{Even no. of A} \\ A & \text{Odd no. of A} \end{cases}$$





2 Min Summary

XOR GATE

Questions

Discussion

Thank You GW Soldiers!

